Weak-to-Strong Flow Direct-OPD

Transferring a small model's RL-induced policy shift to a larger flow model

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Abstract

Suppose reinforcement learning is affordable for a small flow model but expensive for a much larger model. Let S_0 be the small base checkpoint, S_R the same model after RL, and L_0 a larger base model that we wish to improve. Directly distilling S_R into the large model is undesirable: it transfers both the useful RL change and the small model's capacity limitations. Motivated by Direct On-Policy Distillation for language models, we instead transfer the *policy shift* from S_0 to S_R and apply it on states visited by the large recipient.

The formulation does not require flow models themselves to be Gaussian. At the most general level, the donor signal is the Radon–Nikodym ratio $\mathrm{d}K_{S_R}/\mathrm{d}K_{S_0}$ between two conditional Markov kernels, and the exact KL-regularized target is the geometric tilt

$$K_{L_0}\left(\frac{\mathrm{d}K_{S_R}}{\mathrm{d}K_{S_0}}\right)^{\lambda}.$$

Gaussianity enters only when a stochastic sampler is chosen to make this target tractable. For continuous SDEs with shared diffusion, Girsanov's theorem turns the local donor shift into a drift correction $b_{L_0} + \lambda(b_{S_R} - b_{S_0})$; the globally tilted path law generally requires an additional Doob continuation-value correction. For Euler SDE and SDE-DPM-Solver++ kernels, the conditional target is Gaussian and the mean shift is available in closed form. For deterministic ODE sampling, transition likelihood ratios are singular, but the same velocity-delta target survives as a rescaled small-noise control-energy limit. We derive all four levels, give a recipient-on-policy training algorithm, and state compatibility checks and experiments for FLUX.2-klein-base-9B $\rightarrow$ FLUX.2-dev.

Contents

1 Motivation, roles, and literature

1.1 The weak-to-strong setting

We use role-based notation rather than naming models by size:

symbol	role	concrete example
S_0	small donor reference before RL	FLUX.2-klein-base-9B
S_R	same donor after task RL	a GenEval2 Flow-DPPO 9B LoRA
L_0	large recipient reference	FLUX.2-dev, approximately 32B
L_θ	trainable large recipient	initialized from L_0
$L_{\bar{\theta}}$	frozen rollout behavior checkpoint	refreshed from L_θ

The small pair is a *donor* and the large model is a *recipient*. Calling the 9B checkpoint "teacher" and the 32B checkpoint "student" obscures the central point: the final 9B policy need not be better than the 32B model. The transferable object is the change induced by RL,

$$\Delta v_S(t, x, c) := v_{S_R}(t, x, c) - v_{S_0}(t, x, c), \tag{1}$$

not the absolute field v_{S_R}.

Directly matching v_{L_θ} to v_{S_R} imports the donor's pretraining biases, capacity ceiling, and any missing knowledge. Base subtraction removes what the donor already preferred before RL and isolates what RL changed.

1.2 The matching language-model literature

The same idea has appeared in increasingly direct forms:

- Proxy-Tuning [2] adds $\text{logits}_{\text{small,tuned}} - \text{logits}_{\text{small,base}}$ to a larger base model at decoding time.
- RAST [3] specializes the donor change to an RL-induced reasoning shift and tests transfer across model scales.
- THINKLOGIT [4] uses the same arithmetic for long-reasoning transfer and optionally trains the small guider against target-aware preferences.
- Logit Grafting [5] shows that additive post-training logit deltas are exponential tilts and the exact solution of a KL-regularized reward objective.
- Direct-OPD [1], posted to arXiv in July 2026, is the closest match to the present goal. It treats $\log \pi_{\text{small,RL}} - \log \pi_{\text{small,ref}}$ as a dense implicit reward and trains a stronger student on its own on-policy states.

Categorical logits are natural parameters of the next-token distribution, so adding a logit delta gives an exact exponential tilt. A flow velocity is not generally a log density or a natural parameter. The remainder of this note identifies the conditions under which velocity arithmetic has the same exact interpretation and what remains when those conditions do not hold.

Scope

The proposed method transfers an observed RL-induced policy shift. Calling that shift the "original RL reward" requires the additional approximation that S_R is near the optimum of a KL-regularized RL problem anchored at S_0. GRPO clipping, reward normalization, finite optimization, or zero-KL RL weaken that literal reward interpretation, but do not invalidate the checkpoint-pair policy shift itself.

2 General non-Gaussian kernel formulation

2.1 Recipient-on-policy query states

Let

$$z = (t, x, c) \tag{2}$$

denote a physical time, latent state, and raw conditioning input. At outer iteration k, we sample z from a frozen recipient behavior process $L_{\bar{\theta}}$. All donor and recipient models are then queried on the same physical (t, x, c), although each architecture may encode the raw condition c with its own text encoder.

For the moment, let

$$K_j(\mathrm{d}y \mid z), \qquad j \in \{S_0, S_R, L_0, L_\theta\}, \tag{3}$$

be arbitrary conditional Markov kernels on the same next-state space. They need not be Gaussian. Assume

$$K_{S_R}(\cdot \mid z) \ll K_{S_0}(\cdot \mid z), \qquad K_{L_0}(\cdot \mid z) \ll K_{S_0}(\cdot \mid z), \tag{4}$$

so a version of the donor ratio is defined $K_{L_0}(\cdot \mid z)$-almost everywhere.

Definition 2.1 (Donor policy shift)**.** The small-model RL shift at query z is the Radon–Nikodym derivative

$$R_S(y \mid z) := \frac{\mathrm{d}K_{S_R}(\cdot \mid z)}{\mathrm{d}K_{S_0}(\cdot \mid z)}(y), \qquad r_S(y \mid z) := \log R_S(y \mid z). \tag{5}$$

The ratio is larger than one on next states whose probability RL increased and smaller than one where RL suppressed probability. It removes the donor's absolute base policy.

2.2 Exact geometric tilt

For transfer strength $\lambda > 0$, define the conditional normalizer

$$Z_\lambda(z) := \int R_S(y \mid z)^\lambda K_{L_0}(\mathrm{d}y \mid z). \tag{6}$$

For $\lambda = 0$, define $R_S^0 \equiv 1$, so $Z_0 = 1$ and $Q_0 = K_{L_0}$ even on sets where $R_S = 0$.

Proposition 2.2 (KL-regularized donor-shift target)**.** *For $\lambda > 0$, if $0 < Z_\lambda(z) < \infty$, the kernel*

$$\frac{\mathrm{d}Q_\lambda(\cdot \mid z)}{\mathrm{d}K_{L_0}(\cdot \mid z)}(y) = \frac{R_S(y \mid z)^\lambda}{Z_\lambda(z)} \tag{7}$$

is the unique maximizer of

$$\arg\max_K \left\{\lambda \mathbb{E}_{Y\sim K}[r_S(Y \mid z)] - \mathrm{KL}(K \| K_{L_0})\right\}. \tag{8}$$

Equivalently, for every $K \ll K_{L_0}$ for which the two sides are well-defined in the extended-real sense,

$$\lambda \mathbb{E}_K[r_S] - \mathrm{KL}(K \| K_{L_0}) = \log Z_\lambda - \mathrm{KL}(K \| Q_\lambda). \tag{9}$$

If $K \not\ll Q_\lambda$, the right side and the variational objective are both interpreted as $-\infty$ rather than as an indeterminate difference of infinities.

Proof. From (7),

$$\log \frac{\mathrm{d}Q_\lambda}{\mathrm{d}K_{L_0}} = \lambda r_S - \log Z_\lambda.$$

Substitute this identity into $\mathrm{KL}(K \| Q_\lambda) = \mathbb{E}_K[\log(\mathrm{d}K/\mathrm{d}Q_\lambda)]$ and rearrange to obtain (9). Nonnegativity of KL gives the maximizer. □

For densities with respect to a common base measure,

$$q_\lambda(y \mid z) = \frac{k_{L_0}(y \mid z)\,[k_{S_R}(y \mid z)/k_{S_0}(y \mid z)]^\lambda}{Z_\lambda(z)}. \tag{10}$$

This is a *geometric* or product-of-experts tilt. It must not be confused with the arithmetic probability mixture used by P-OPD,

$$(1-\alpha)K_{\mathrm{old}} + \alpha K_T. \tag{11}$$

The geometric target has no component-responsibility sigmoid.

2.3 Implicit and non-Gaussian kernels

The general construction is useful beyond Gaussian transitions whenever the ratio can be estimated:

1. If all log densities are evaluable, optimize

   $$\mathrm{KL}(K_{L_\theta} \| K_{L_0}) - \lambda \mathbb{E}_{K_{L_\theta}}[r_S]. \tag{12}$$

   The unknown $\log Z_\lambda$ is independent of the trainable parameters.

2. If only donor transition samples are available, train a conditional discriminator with equal class priors. Its Bayes optimum obeys

   $$D^*(z,y) = \frac{k_{S_R}(y \mid z)}{k_{S_R}(y \mid z) + k_{S_0}(y \mid z)}, \qquad \operatorname{logit} D^*(z,y) = r_S(y \mid z). \tag{13}$$

   Both classes must use the same query-state distribution; otherwise the classifier also learns a state-occupancy ratio.

3. One may sample approximately from Q_λ by weighting K_{L_0} transitions with R_S^λ, followed by resampling or SMC. This is exact in the population limit but can have severe effective-sample-size collapse.

4. MMD, OT, score matching, Stein losses, or conditional flow matching can project implicit kernels into a trainable model. Without an exact ratio identity they are surrogates, not exact realizations of (7).

   The kernel formulation therefore separates two statements:

- *Flow matching does not require Gaussian probability paths.*
- *A chosen stochastic sampler may use Gaussian increments because they make the policy-shift ratio and target loss tractable.*

3 Continuous-SDE formulation

3.1 Shared-diffusion processes and Girsanov ratio

Use an increasing-time coordinate $\tau \in [0,T]$ and consider

$$\mathrm{d}X_\tau = b_j(\tau, X_\tau, c)\mathrm{d}\tau + \sigma_\tau \mathrm{d}W_\tau, \qquad a_\tau := \sigma_\tau\sigma_\tau^\top \succ 0. \tag{14}$$

All donor and recipient processes share the diffusion coefficient and initial law; only their drifts differ. We assume the required Girsanov exponentials are true martingales for every pair whose path measures are compared. Define

$$\Delta b_S := b_{S_R} - b_{S_0}, \qquad u_\tau := \sigma_\tau^\top a_\tau^{-1} \Delta b_S. \tag{15}$$

Proposition 3.1 (Donor path ratio)**.** *Under a Novikov or equivalent martingale condition, Girsanov's theorem gives*

$$\log \frac{\mathrm{d}P_{S_R}}{\mathrm{d}P_{S_0}} = \int_0^T u_\tau^\top \mathrm{d}W_\tau^{S_0} - \frac{1}{2}\int_0^T \|u_\tau\|^2 \,\mathrm{d}\tau \tag{16}$$

$$= \int_0^T \Delta b_S^\top a_\tau^{-1} \left[\mathrm{d}X_\tau - \frac{b_{S_R} + b_{S_0}}{2}\mathrm{d}\tau\right]. \tag{17}$$

All fields are evaluated along the same path $X_{0:T}$.

The corresponding path KL between two shared-diffusion processes is

$$\mathrm{KL}(P_b \| P_{L_0}) = \frac{1}{2}\mathbb{E}_{P_b}\int_0^T \|b - b_{L_0}\|^2_{a_\tau^{-1}} \,\mathrm{d}\tau. \tag{18}$$

This is an exact stochastic-process identity, not a Gaussian one-step approximation [10].

3.2 Local zero-discount control target

At a frozen (τ, x, c), consider using a candidate drift b over one infinitesimal interval. The expected donor log-ratio rate under that candidate is

$$\mathfrak{r}_S(b) = \Delta b_S^\top a^{-1}(b - b_{S_0}) - \frac{1}{2}\|\Delta b_S\|^2_{a^{-1}}. \tag{19}$$

The recipient drift-KL rate relative to L_0 is

$$\mathfrak{c}_{L_0}(b) = \frac{1}{2}\|b - b_{L_0}\|^2_{a^{-1}}. \tag{20}$$

Proposition 3.2 (Local SDE policy-shift graft)**.** *The frozen-state objective*

$$\mathcal{H}_\lambda(b) = \lambda \mathfrak{r}_S(b) - \mathfrak{c}_{L_0}(b) \tag{21}$$

has the unique optimum

$$\boxed{b^* = b_{L_0} + \lambda(b_{S_R} - b_{S_0}).} \tag{22}$$

Proof. Terms independent of b can be discarded. Completing the square gives

$$\mathcal{H}_\lambda(b) = \text{constant} - \frac{1}{2}\|b - [b_{L_0} + \lambda \Delta b_S]\|^2_{a^{-1}}.$$

Positive definiteness of a gives the unique optimum. □

Equation (22) is the continuous-SDE counterpart of adding a small-model logit delta to a large language model. It is exact for the frozen-state, immediate-reward problem. Using recipient on-policy states but detaching them during optimization produces a semi-gradient version of this objective.

3.3 Why the globally tilted path needs a value correction

For this subsection, additionally assume

$$P_{L_0} \ll P_{S_0}, \qquad 0 < \mathcal{Z}_\lambda := \mathbb{E}_{P_{L_0}}\left[\left(\frac{\mathrm{d}P_{S_R}}{\mathrm{d}P_{S_0}}\right)^\lambda\right] < \infty. \tag{23}$$

The ideal path-space target is

$$\frac{\mathrm{d}\mathbb{Q}_\lambda}{\mathrm{d}P_{L_0}} = \frac{1}{\mathcal{Z}_\lambda}\left(\frac{\mathrm{d}P_{S_R}}{\mathrm{d}P_{S_0}}\right)^\lambda. \tag{24}$$

Define

$$\bar{b} = b_{L_0} + \lambda \Delta b_S \tag{25}$$

and the state-dependent potential

$$U_\lambda = \lambda \Delta b_S^\top a^{-1}(b_{L_0} - b_{S_0}) + \frac{\lambda(\lambda-1)}{2} \|\Delta b_S\|_{a^{-1}}^2 . \tag{26}$$

Completing the stochastic exponential shows that (24) is a Feynman–Kac weighting of the process with drift $\bar{b}$. Let

$$H(\tau, x) = \mathbb{E}_{\bar{b}}^{\tau,x}\left[\exp\left(\int_\tau^T U_\lambda(s, X_s)\mathrm{d}s\right)\right]. \tag{27}$$

Under the usual regularity conditions, H solves

$$\partial_\tau H + \bar{b} \cdot \nabla H + \frac{1}{2} a : \nabla^2 H + U_\lambda H = 0, \qquad H(T, x) = 1. \tag{28}$$

The globally tilted process has Doob-transformed drift

$$\boxed{b_{\mathbb{Q}_\lambda} = b_{L_0} + \lambda(b_{S_R} - b_{S_0}) + a\nabla \log H.} \tag{29}$$

If the shared initial law is ρ_0, the unconditional global tilt also changes it:

$$\mathbb{Q}_\lambda(X_0 \in \mathrm{d}x) = \frac{H(0, x)\rho_0(\mathrm{d}x)}{\mathcal{Z}_\lambda}, \qquad \mathcal{Z}_\lambda = \int H(0, x)\rho_0(\mathrm{d}x). \tag{30}$$

Therefore the drift (29) started from the original ρ_0 does not by itself realize the unconditional target (24). If the generative base-noise law must remain fixed, define the tilt conditionally on $X_0 = x$ and normalize each conditional path law by $H(0, x)$ before averaging over the original ρ_0.

The last term is a continuation-value correction. It vanishes only in special cases, for example when U_λ is spatially constant. Therefore simple drift arithmetic should be described as the local, zero-discount Direct-OPD target, not as the exact globally normalized path tilt. This is the continuous analogue of Direct-OPD's practical immediate-token surrogate versus its idealized sequence-level objective.

4 Gaussian, Euler-SDE, and DPM-SDE specializations

4.1 Generic shared-covariance Gaussian transition

Condition on a recipient query state z. Suppose the donor pair shares covariance Σ_S and the recipient pair shares covariance Σ_L:

$$\begin{aligned} K_{S_0} &= \mathcal{N}(\mu_{S_0}, \Sigma_S), & K_{S_R} &= \mathcal{N}(\mu_{S_R}, \Sigma_S), \\ K_{L_0} &= \mathcal{N}(\mu_{L_0}, \Sigma_L), & K_{L_\theta} &= \mathcal{N}(\mu_\theta, \Sigma_L). \end{aligned} \tag{31}$$

Define

$$\Delta\mu_S := \mu_{S_R} - \mu_{S_0}, \qquad \bar{\mu}_S := \frac{\mu_{S_R} + \mu_{S_0}}{2}. \tag{32}$$

The donor log ratio is linear in y:

$$r_S(y) = \Delta\mu_S^\top \Sigma_S^{-1}(y - \bar{\mu}_S). \tag{33}$$

Proposition 4.1 (Closed-form Gaussian geometric tilt)**.** *The kernel target* (7) *is*

$$Q_\lambda = \mathcal{N}(\mu_\lambda^*, \Sigma_L), \qquad \boxed{\mu_\lambda^* = \mu_{L_0} + \lambda \Sigma_L \Sigma_S^{-1} \Delta\mu_S.} \tag{34}$$

Consequently,

$$\boxed{\mathrm{KL}(K_{L_\theta} \| Q_\lambda) = \frac{1}{2} \left\| \mu_\theta - \mu_{L_0} - \lambda \Sigma_L \Sigma_S^{-1} \Delta\mu_S \right\|^2_{\Sigma_L^{-1}} .} \tag{35}$$

Proof. Insert (33) into $\log K_{L_0}(y) + \lambda r_S(y)$. Its quadratic term remains $-\frac{1}{2} y^\top \Sigma_L^{-1} y$ and its linear term is

$$y^\top \left[\Sigma_L^{-1} \mu_{L_0} + \lambda \Sigma_S^{-1} \Delta\mu_S \right].$$

Completing the square gives (34). The equal-covariance Gaussian KL from K_{L_θ} to Q_λ gives (35). □

If all four covariances are the same,

$$\boxed{\mu_\lambda^* = \mu_{L_0} + \lambda(\mu_{S_R} - \mu_{S_0}).} \tag{36}$$

Thus mean-delta arithmetic is not an arbitrary heuristic in this regime; it is the exact natural parameter update selected by the general kernel objective.

Remark 4.2 (Unequal donor covariances)**.** If S_0 and S_R do not share covariance, the donor reward is quadratic and the target precision becomes

$$\Lambda_* = \Lambda_{L_0} + \lambda(\Lambda_{S_R} - \Lambda_{S_0}), \tag{37}$$

which is normalizable only if $\Lambda_* \succ 0$. Plain mean or velocity subtraction is then not the exact target.

4.2 Euler–Maruyama SDE

For a step of length $h > 0$, let

$$K_j^h(\cdot \mid x) = \mathcal{N}(x + h\, b_j,\ hA), \qquad A \succ 0. \tag{38}$$

All drifts are evaluated at the same (t, x, c). Proposition 4.1 gives

$$Q_\lambda^h = \mathcal{N}(x + h[b_{L_0} + \lambda(b_{S_R} - b_{S_0})], hA). \tag{39}$$

Writing

$$b_* := b_{L_0} + \lambda(b_{S_R} - b_{S_0}), \tag{40}$$

the exact event-sum conditional KL is

$$\ell_{\mathrm{Euler,sum}} = \frac{h}{2} \| b_{L_\theta} - b_* \|^2_{A^{-1}} . \tag{41}$$

For isotropic $A = g_t^2 I$ and event dimension D,

$$\ell_{\mathrm{Euler,sum}} = \frac{h}{2g_t^2} \sum_{d=1}^{D} (b_{L_\theta,d} - b_{*,d})^2, \tag{42}$$

$$\ell_{\mathrm{Euler,mean}} = \frac{1}{D} \ell_{\mathrm{Euler,sum}}. \tag{43}$$

The sum is the joint transition KL; the mean is its dimension-normalized version. They have the same isolated minimizer but different strength relative to other losses and trust budgets.

4.3 First-order SDE-DPM-Solver++

SDE-DPM-Solver++ is a stochastic solver for a *chosen Gaussian diffusion path*; it is not a generic property of arbitrary flow matching. Assume

$$X_t = \alpha_t X_0 + \sigma_t \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \qquad \ell_t := \log \frac{\alpha_t}{\sigma_t}. \tag{44}$$

For a reverse step from noisier s to cleaner t, set

$$h := \ell_t - \ell_s > 0. \tag{45}$$

The first-order SDE-DPM-Solver++ update [7] is

$$X_t = \underbrace{\frac{\sigma_t}{\sigma_s} e^{-h}}_{A_{t,s}} X_s + \underbrace{\alpha_t (1 - e^{-2h})}_{B_{t,s}} \widehat{x}_{0,j}(X_s, s) + \underbrace{\sigma_t \sqrt{1 - e^{-2h}}}_{C_{t,s}} \xi, \qquad \xi \sim \mathcal{N}(0, I). \tag{46}$$

Conditioned on $X_s = x_s$,

$$K_j(\cdot \mid x_s) = \mathcal{N}(\mu_j, \Sigma_{t,s}), \quad \mu_j = A_{t,s} x_s + B_{t,s} \widehat{x}_{0,j}, \quad \Sigma_{t,s} = C_{t,s}^2 I. \tag{47}$$

The stochastic covariance is model independent and the mean is affine in the model's data prediction. Therefore the exact graft is

$$\begin{aligned} \mu_* &= \mu_{L_0} + \lambda(\mu_{S_R} - \mu_{S_0}) \\ &= A_{t,s} x_s + B_{t,s} \left[\widehat{x}_{0,L_0} + \lambda(\widehat{x}_{0,S_R} - \widehat{x}_{0,S_0}) \right], \end{aligned} \tag{48}$$

or equivalently

$$\boxed{\widehat{x}_{0,*} = \widehat{x}_{0,L_0} + \lambda(\widehat{x}_{0,S_R} - \widehat{x}_{0,S_0}).} \tag{49}$$

The exact solver-kernel loss is

$$\mathrm{KL}(K_{L_\theta} \| Q_\lambda) = \frac{B_{t,s}^2}{2 C_{t,s}^2} \left\| \widehat{x}_{0,L_\theta} - \widehat{x}_{0,*} \right\|^2. \tag{50}$$

This equality is exact for the selected first-order numerical transition kernel. It does not claim that the numerical solver exactly integrates the underlying continuous SDE.

4.4 Multistep SDE-DPM-Solver++

A multistep solver is Markov only after its history is included in the state. For SDE-DPM-Solver++(2M), let r be the previous history point and define

$$r_0 := \frac{\ell_s - \ell_r}{h} > 0, \quad D_{0,j} := \widehat{x}_{0,j}(x_s, s), \quad D_{1,j} := \frac{\widehat{x}_{0,j}(x_s, s) - \widehat{x}_{0,j}(x_r, r)}{r_0}. \tag{51}$$

Equation (20) of DPM-Solver++ can be written

$$\mu_j = A_{t,s} x_s + B_{t,s} D_{0,j} + \frac{B_{t,s}}{2} D_{1,j}, \qquad \Sigma_{t,s} = C_{t,s}^2 I. \tag{52}$$

The exact target conditioned on the common augmented history is

$$\boxed{\mu_* = \mu_{L_0} + \lambda(\mu_{S_R} - \mu_{S_0}).} \tag{53}$$

Every current and historical model-output term contributes to the mean difference. Grafting only the current velocity or current $\widehat{x}_0$ omits the multistep derivative correction and is not the exact conditional target.

More generally, if

$$\mu_j(\mathcal{H}_n) = A_n x_n + \sum_{q=0}^{m-1} B_{nq} M_{j,nq}(\mathcal{H}_n), \tag{54}$$

then compute the *whole solver mean* for each model on the same history and graft $\mu_{L_0} + \lambda(\mu_{S_R} - \mu_{S_0})$. Marginalizing random histories generally produces a non-Gaussian mixture kernel.

4.5 When a flow velocity can use DPM-SDE

For a known affine Gaussian conditional path

$$X_t = \alpha_t X_0 + \sigma_t \varepsilon, \qquad \dot{X}_t = \dot{\alpha}_t X_0 + \dot{\sigma}_t \varepsilon, \tag{55}$$

define

$$\mathcal{D}_t := \alpha_t \dot{\sigma}_t - \sigma_t \dot{\alpha}_t \neq 0. \tag{56}$$

A flow-velocity prediction $v_j(t, x)$ converts to

$$\widehat{x}_{0,j} = \frac{\dot{\sigma}_t x - \sigma_t v_j(t,x)}{\mathcal{D}_t}, \tag{57}$$

$$\widehat{\varepsilon}_j = \frac{-\dot{\alpha}_t x + \alpha_t v_j(t,x)}{\mathcal{D}_t}. \tag{58}$$

For normalized VP schedules satisfying $\alpha_t^2 + \sigma_t^2 = 1$, standard diffusion v-prediction defines

$$v_{\text{pred}} = \alpha_t \varepsilon - \sigma_t X_0, \qquad \widehat{x}_0 = \alpha_t x - \sigma_t v_{\text{pred}}. \tag{59}$$

The physical path derivative and the standard network output differ by

$$v_{\text{physical}} = \mathcal{D}_t \, v_{\text{pred}}. \tag{60}$$

Thus a raw v_{pred} must not be substituted for the physical flow velocity without this schedule-dependent conversion.

DPM-SDE is applicable only when the flow model's parameterization belongs to a compatible Gaussian diffusion schedule with monotone half-logSNR and a valid reverse SDE. An arbitrary non-Gaussian flow-matching path, nonlinear interpolation, or bare learned ODE field does not canonically define $\widehat{x}_0$, a score, a reverse diffusion, or a DPM-SDE transition.

DPM-SDE exactness boundary

The final zero-variance DPM step is deterministic and has no transition-density ratio. Exact kernel grafting requires positive model-independent covariance, a common augmented solver history, and model outputs converted into the same prediction parameterization.

5 Deterministic ODE formulations

5.1 Why transition ratios fail

For an ODE Euler step,

$$K_j(\mathrm{d}y \mid x) = \delta_{x + h\, v_j(t,x,c)}(\mathrm{d}y). \tag{61}$$

If the donor maps differ, K_{S_R} and K_{S_0} are mutually singular, so $\mathrm{d}K_{S_R}/\mathrm{d}K_{S_0}$ is undefined. Girsanov also fails because the diffusion covariance is zero. No deterministic transition KL or policy ratio can justify the target directly.

5.2 Selected small-noise control limit

Introduce a reference small-noise family

$$K_j^{\varepsilon,h} = \mathcal{N}(x + h\, v_j,\ \varepsilon h A), \qquad A \succ 0. \tag{62}$$

For every $\varepsilon > 0$, the exact geometric target has mean

$$x + h\left[v_{L_0} + \lambda(v_{S_R} - v_{S_0})\right]. \tag{63}$$

Define

$$v_* := v_{L_0} + \lambda(v_{S_R} - v_{S_0}). \tag{64}$$

Then

$$\frac{\varepsilon}{h}\mathrm{KL}(K^{\varepsilon,h}_{L_\theta} \| Q^{\varepsilon,h}_\lambda) = \frac{1}{2}\left\| v_{L_\theta} - v_* \right\|^2_{A^{-1}}. \tag{65}$$

The ratio itself does not exist at $\varepsilon = 0$, but the Gaussian family selects a finite deterministic target and its rescaled KL is exactly a quadratic control metric.

The practical recipient-on-policy ODE loss is therefore

$$\boxed{\mathcal{L}_{\mathrm{ODE}} = \mathbb{E}_{\substack{C,t \\ X_t \sim \rho_{L_{\bar\theta},t}}}\left[w(t) \left\| [v_{L_\theta} - v_{L_0}] - \lambda[v_{S_R} - v_{S_0}] \right\|^2_{G_t} \right].} \tag{66}$$

Without a declared small-noise SDE, G_t and $w(t)$ are modeling choices. Common options are identity velocity geometry, a scheduler-derived control metric, or a per-dimension RMS normalization.

5.3 Lagrangian flow-map delta

The Eulerian field loss queries the 9B donor on 32B recipient states. If those states are too far outside the donor occupancy, a Lagrangian alternative evaluates every model on its own trajectory. Let $\Phi_{j,t}(Z, c)$ be flow maps from common base noise Z and define

$$\Psi^\lambda_t(Z, c) = \Phi_{L_0,t}(Z, c) + \lambda \left[\Phi_{S_R,t}(Z, c) - \Phi_{S_0,t}(Z, c) \right]. \tag{67}$$

Its path derivative is

$$\begin{aligned} \dot\Psi^\lambda_t &= v_{L_0}(t, \Phi_{L_0,t}, c) \\ &\quad + \lambda \left[v_{S_R}(t, \Phi_{S_R,t}, c) - v_{S_0}(t, \Phi_{S_0,t}, c) \right]. \end{aligned} \tag{68}$$

Conditional flow matching uses

$$\mathcal{L}_{\text{map-CFM}} = \mathbb{E}_{C,Z,t} \left\| v_\theta(t, \Psi^\lambda_t, C) - \dot\Psi^\lambda_t \right\|^2. \tag{69}$$

Its population optimum is the conditional mean

$$u_t(x, c) = \mathbb{E}[\dot\Psi^\lambda_t(Z, c) \mid \Psi^\lambda_t(Z, c) = x]. \tag{70}$$

Under standard regularity, this field transports the chosen pseudo-path marginals [6]. It is exact for the explicitly declared common-noise coupling, not a zero-noise policy-ratio identity.

5.4 Marginal density-ratio target

ODE time marginals may possess smooth densities even though their transition kernels are Dirac. Assume, at each (t, c),

$$\rho_{S_R,t}(\cdot \mid c) \ll \rho_{S_0,t}(\cdot \mid c), \qquad \rho_{L_0,t}(\cdot \mid c) \ll \rho_{S_0,t}(\cdot \mid c), \tag{71}$$

and define the finite positive normalizer

$$Z_t(c) = \int \rho_{L_0,t}(x \mid c) \left[\frac{\rho_{S_R,t}(x \mid c)}{\rho_{S_0,t}(x \mid c)} \right]^\lambda \mathrm{d}x \in (0, \infty). \tag{72}$$

The static density target is

$$q_t(x \mid c) = \frac{1}{Z_t(c)} \rho_{L_0,t}(x \mid c) \left[\frac{\rho_{S_R,t}(x \mid c)}{\rho_{S_0,t}(x \mid c)} \right]^\lambda. \tag{73}$$

On the region where all three component densities are positive, its static score identity is

$$\nabla \log q_t = \nabla \log \rho_{L_0,t} + \lambda \left[\nabla \log \rho_{S_R,t} - \nabla \log \rho_{S_0,t}\right]. \tag{74}$$

However, products and ratios of densities following their respective continuity or Fokker–Planck dynamics do not generally follow the dynamics induced by score arithmetic. Consequently, score or velocity arithmetic does not automatically generate the complete path (73). A strict endpoint version requires an endpoint density-ratio estimator, importance resampling, DMD with an online fake score, or a dynamic corrector. This is a separate and more expensive distribution-level method.

6 Main training objective and algorithm

6.1 Recommended transition-mean implementation

The safest implementation uses the *actual scheduler or solver transition mean* rather than assuming a particular Euler coefficient. On a recipient behavior state and common solver history, compute

$$\Delta\mu_S = \mu_{S_R} - \mu_{S_0}. \tag{75}$$

When donor and recipient covariances differ but each pair is internally matched, define

$$d_\mu := \Sigma_L \Sigma_S^{-1} \Delta\mu_S. \tag{76}$$

The unit-strength recipient trust cost is

$$K_1 := \frac{1}{2} d_\mu^\top \Sigma_L^{-1} d_\mu. \tag{77}$$

Define the corresponding per-dimension quantities

$$\bar{K}_1 := K_1/D, \qquad \bar{\kappa} := \kappa/D. \tag{78}$$

Given nominal strength λ_0 and either a conditional joint-KL budget κ or the equivalent per-dimension budget $\bar{\kappa}$, set

$$\lambda_{\text{eff}} = \begin{cases} \lambda_0, & K_1 = 0, \\ \min\left\{\lambda_0, \sqrt{\kappa/K_1}\right\} = \min\left\{\lambda_0, \sqrt{\bar{\kappa}/\bar{K}_1}\right\}, & K_1 > 0. \end{cases} \tag{79}$$

This guarantees

$$\mathrm{KL}(Q_{\lambda_{\text{eff}}} \| K_{L_0}) \le \kappa. \tag{80}$$

The target mean is

$$\mu_{\text{target}} = \mu_{L_0} + \lambda_{\text{eff}} d_\mu. \tag{81}$$

The dimension-normalized training loss is

$$\boxed{\mathcal{L}_{\text{Flow-Direct-OPD}} = \mathbb{E}_{z\sim\nu_{L_{\bar{\theta}}}}\left[\frac{1}{2D}\left\|\mu_\theta(z) - \mathrm{sg}(\mu_{\text{target}}(z))\right\|^2_{\Sigma_L^{-1}}\right].} \tag{82}$$

Removing $1/D$ recovers the exact joint conditional KL. The displayed optimization loss uses the per-dimension convention; (80) is the equivalent joint guarantee. Equation (78) makes the conversion explicit.

If one shared isotropic Euler-SDE wrapper is used, this simplifies to

$$v_{\text{target}} = v_{L_0} + \lambda_{\text{eff}}(v_{S_R} - v_{S_0}), \tag{83}$$

$$\mathcal{L} = \mathbb{E}_{\nu_{L_{\bar{\theta}}}}\left[\frac{h_t}{2Dg_t^2}\left\|v_{L_\theta} - \mathrm{sg}(v_{\text{target}})\right\|^2\right]. \tag{84}$$

For the ODE arm, use (66) instead of claiming a transition KL.

6.2 Recipient-on-policy optimization

Algorithm 1 Weak-to-Strong Flow Direct-OPD

Require: frozen donor reference S_0; frozen donor RL checkpoint S_R
Require: large recipient reference L_0; trainable recipient $L_\theta \leftarrow L_0$
Require: nominal strength λ_0; optional trust budget κ
for outer iteration $k = 0, 1, \dots$ **do**
  Freeze behavior checkpoint $L_{\bar{\theta}} \leftarrow L_\theta$
  Roll out $L_{\bar{\theta}}$ and cache recipient states, physical times, and solver histories
  for each sampled training state/history z **do**
    Evaluate $\mu_{S_0}(z)$ and $\mu_{S_R}(z)$ without gradients
    Evaluate $\mu_{L_0}(z)$ without gradients
    Compute $\Delta\mu_S$, d_μ, and K_1
    If κ is absent set $\lambda_{\text{eff}} \leftarrow \lambda_0$; otherwise use (79)
    Set $\mu_{\text{target}} \leftarrow \mu_{L_0} + \lambda_{\text{eff}} d_\mu$
    Evaluate trainable $\mu_\theta(z)$ and accumulate (82)
  end for
  Update θ; log realized recipient-to-base displacement and refresh behavior
end for

The donor pair and recipient reference are detached. The query distribution is the recipient's own behavior occupancy, which is the flow analogue of Direct-OPD evaluating the weak checkpoint ratio on strong-student prefixes. Refreshing the behavior model limits stale-state error, but the detached optimization remains a local semi-gradient rather than the full path gradient of (24).

6.3 Reliability masks and composition

If the donor shift is unreliable on some recipient states, introduce a detached mask $m(z) \in [0, 1]$:

$$\mu_{\text{target}} = \mu_{L_0} + m(z)\lambda_{\text{eff}} d_\mu. \tag{85}$$

Possible masks use cross-seed or cross-checkpoint donor-delta agreement, donor-state OOD tests, or explicit domain routing. A mask based only on $\|\Delta\mu_S\|$ is not a correctness test: a large shift can encode a reward shortcut.

Several independently learned shifts can be composed:

$$\mu_{\text{target}} = \mu_{L_0} + \sum_{m=1}^{M} \lambda_m d_\mu^{(m)}. \tag{86}$$

This is exact for a product of conditional donor ratios when the normalizer exists. In practice, joint trust projection is needed because individually safe shifts can add constructively or conflict.

7 Alternative objectives and their exactness

arm	construction	exact meaning	principal boundary
A	inference-time velocity graft	a declared ODE field; exact conditional Gaussian mean under shared SDE covariance	requires all three frozen models at inference and can move donor queries off-support
B	recipient-on-policy residual field distillation	weighted L^2 projection of arm A under the behavior occupancy	finite capacity and stale occupancy; not an endpoint distribution objective
C	kernel or SDE transition tilt	exact local KL-regularized target (7)	needs density ratio or shared stochastic kernel; global path target has Doob correction
D	endpoint ratio tilt	exact static distribution target (73)	expensive ratio/score estimation; product path is not automatically dynamically consistent
E	common-noise flow-map CFM	exact CFM target for the explicit pseudo-path (67)	coupling- and coordinate-dependent; additive maps can fold or leave the valid region

7.1 Zero-training inference oracle

Before any recipient training, integrate

$$\dot{X}_t = v_{L_0}(t, X_t, c) + \lambda \left[v_{S_R}(t, X_t, c) - v_{S_0}(t, X_t, c) \right]. \tag{87}$$

This needs one 32B and two 9B forwards per solver query, but directly tests whether the field direction transfers. If a correctly implemented closed-loop graft provides no reward–quality Pareto improvement, training arm B toward the same field target is unlikely to rescue the hypothesis.

7.2 Endpoint importance-CFM

Under the support and normalizability assumptions (71)–(72) at $t = 1$, estimate

$$r_1(y, c) = \log \rho_{S_R,1}(y \mid c) - \log \rho_{S_0,1}(y \mid c). \tag{88}$$

Generate $Y_i \sim \rho_{L_0,1}$ and weight

$$w_i = \exp(\lambda r_1(Y_i, c)). \tag{89}$$

The population weight normalizer is $Z_1(c) = \mathbb{E}_{Y \sim \rho_{L_0,1}}[w(Y, c)] \in (0, \infty)$. Resampling by w_i followed by ordinary endpoint CFM targets

$$q_1(y \mid c) \propto \rho_{L_0,1}(y \mid c) e^{\lambda r_1(y,c)} \tag{90}$$

in the population limit. Its feasibility is governed by

$$\mathrm{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{91}$$

Collapsed ESS is an estimation failure, not evidence that policy-shift transfer itself is false. DMD [8] is an alternative when an online fake-score model is affordable.

7.3 Comparison with direct teacher imitation and P-OPD

Vanilla diffusion/flow distillation [9] uses $v_{\text{target}} = v_{S_R}$ and transfers the donor's absolute policy. The canonical case in which it agrees with Flow Direct-OPD is $S_0 = L_0$ and $\lambda = 1$; more generally equality requires $v_{L_0} + \lambda(v_{S_R} - v_{S_0}) = v_{S_R}$ pointwise.

P-OPD uses an arithmetic mixture and a posterior responsibility. Its exact local-gradient interpretation requires a sampled stochastic transition and positive shared covariance; the gate collapses in the deterministic limit. Flow Direct-OPD uses a geometric ratio tilt. Under shared Gaussian covariance the product closes to one Gaussian and no responsibility gate appears. The two methods solve different proximal problems.

8 FLUX.2 compatibility, diagnostics, and experiments

8.1 What the public model configs establish

The public Hugging Face configs for FLUX.2-klein-base-9B and FLUX.2-dev show encouraging structural compatibility:

component	observed relation	implication
scheduler	same FlowMatchEulerDiscreteScheduler fields	physical sigma grids can be matched
transformer I/O	Flux2Transformer2DModel, 128 channels	output tensor geometry can match
RoPE axes	$(32, 32, 32, 32)$ for both	latent token coordinate convention is compatible
VAE config	AutoencoderKLFlux2, 32 latent channels	same declared latent architecture
text encoder	Qwen3 vs. Mistral3	embeddings are not interchangeable
joint attention dim	12288 vs. 15360	each model must use its own conditioning path

Configuration equality is not numerical identity. The two published VAE weight files have different sizes (approximately 168.1MB and 336.2MB), consistent with a precision difference but not proving equal fp32 values. Train-time identity grafting requires direct state and one-step parity tests.

8.2 Fail-closed compatibility checks

Before training:

1. Pin the exact S_0 revision from which the RL checkpoint was produced. Disabling the RL adapter must recover that base numerically.

2. Compare VAE state dictionaries after fp32 conversion; encode the same images with posterior mode and compare packed latent coordinates, latent IDs, scaling, and decode parity.

3. Use one diffusers revision and compare complete sigma/timestep grids, dynamic shift, and solver coefficients at every intended resolution and NFE.

4. Instrument one step of each official pipeline and verify that converted outputs represent the same physical velocity or complete transition mean, including sign.

5. Feed the same raw prompt to each model's own text encoder. Begin with guidance scale 1; equal CFG numbers across the two pipelines need not define equal physical fields.

6. Compute donor subtraction in fp32 and deterministic eval mode. Require its RMS to be well above repeated-forward and dtype noise before interpreting it as an RL signal.

If latent coordinates differ, a state transport map T is required and vector shifts must be pushforwarded by its Jacobian. Matching tensor shape alone is insufficient.

8.3 Required scale and reliability logs

At every sampled timestep, log both RMS and event-L2 scales:

- $\mathrm{RMS}(\Delta v_S)$, $\mathrm{RMS}(v_{L_0})$, and their ratio;
- one-step displacement $|h_t|\,\mathrm{RMS}(\lambda\Delta v_S)$;
- joint and per-dimension K_1, λ_{eff}, and clipping rate;
- open-loop and closed-loop endpoint latent displacement;
- cross-seed, cross-checkpoint, or cross-scale cosine similarity of donor deltas;
- recipient-state OOD diagnostics relative to S_0 and S_R trajectory states;
- realized post-update $\mathrm{RMS}(v_{L_\theta} - v_{L_0})$ versus the requested target.

For donor pairs i, j, let $G_k = A_k^{-1}$ for an SDE control metric, or use the explicitly declared ODE metric. A control-energy weighted cosine is

$$\mathrm{Cos}_{ij} = \frac{\sum_k h_k (\Delta v_{S,k}^{(i)})^\top G_k \Delta v_{S,k}^{(j)}}{\sqrt{\sum_k h_k \left\| \Delta v_{S,k}^{(i)} \right\|_{G_k}^2} \sqrt{\sum_k h_k \left\| \Delta v_{S,k}^{(j)} \right\|_{G_k}^2}}. \tag{92}$$

This measures signal stability, not agreement with an unavailable 32B RL delta.

8.4 Minimum informative experiment sequence

Stage 0: compatibility. Use 32–64 prompts across intended resolutions and timesteps. Stop on latent, scheduler, parameterization, conditioning, or checkpoint-provenance failure.

Stage 1: open-loop field probe. Freeze L_0 trajectories and evaluate all three frozen fields on identical states. Compare the final RL checkpoint with intermediate checkpoints and independent RL seeds. A GenEval-perfect checkpoint may be less transferable than an earlier checkpoint if it encodes reward shortcuts.

Stage 2: closed-loop zero-training oracle. For the deterministic diagnostic only, introduce a signed graft coefficient η and sweep

$$\eta \in \{-0.5, -0.25, 0, 0.125, 0.25, 0.5, 1.0\}. \tag{93}$$

Use η in place of λ in (87). Negative values are sign controls for the ODE field and are outside the one-sided kernel-ratio theory of Section 2; a negative kernel tilt would require mutual absolute continuity and finite negative moments. Include:

- a compute-matched L_0 run with more recipient-only NFEs;
- a $S_0 - S_0$ sham delta;
- prompt-shuffled and norm-matched random deltas;
- absolute S_R and S_0 fields scaled to the same RMS;
- negative η, which should tend to reverse a genuine RL-improvement direction.

Stage 3: recipient training. Only if two neighboring positive strengths improve the reward–quality frontier, train (82). Use the smallest strength within uncertainty of the best oracle result and refresh recipient rollouts frequently.

Stage 4: alternative representations. If field grafting fails because donor queries are OOD on recipient states, test flow-map CFM (69). Attempt endpoint ratio/DMD only after verifying ratio calibration and adequate ESS.

8.5 Evaluation and failure modes

For the GenEval example, report overall and per-category GenEval together with prompt alignment, PickScore, OCR, aesthetics/realism, diversity, duplicate rate, and editing performance. A perfect GenEval score is not sufficient evidence of transferable general capability.

Likely failures include:

- the 9B RL delta encodes detector-specific shortcuts rather than compositional improvement;
- FLUX.2-dev may already have little GenEval headroom, which must be measured under the exact evaluation protocol;
- recipient trajectories leave the donor's reliable state region, especially late in denoising;
- Qwen-conditioned donor corrections are not compatible with Mistral-conditioned recipient behavior;
- the final RL checkpoint overshoots while an intermediate checkpoint transfers;
- a numerically small delta is dominated by bf16 or nondeterministic-forward noise;
- an open-loop-safe direction becomes unstable after closed-loop state drift;
- a fixed global λ spends its scale on timestep or resolution differences;
- field loss improves while reward and perceptual quality degrade, demonstrating objective mismatch rather than implementation success.

Recommended first implementation

Use inference-time ODE velocity grafting as the causal oracle. If it succeeds, distill the recipient residual on recipient on-policy states. Keep Euler-SDE or SDE-DPM-Solver++ as the probabilistically exact transition arm and scale-calibration reference. Do not begin with endpoint density-ratio/DMD: it is more expensive and cannot repair an intrinsically non-transferable donor shift.

8.6 Flow-Factory implementation

The framework implementation uses trainer key `flow-direct-opd` and lives under `src/flow_factory/trainers/fdopd/`. The canonical configuration is `examples/flow_direct_opd/lora/flux2/default.yaml`. It uses the Multi-Reward donor LoRA

`https://huggingface.co/Tencent-Hunyuan-Multimodal-RL/FLUX2-klein-base-9b-GenEval2-Multi-Reward`

by default; replacing only `donor_rl_lora_path` with the corresponding Single-Reward repository gives the first policy-shift ablation.

The implemented V1 dynamics are ODE velocity residuals and the existing Flow-SDE, Dance-SDE, and CPS transition-mean wrappers. DPM-SDE remains a theoretical extension because the current Flow-Factory scheduler registry does not expose a DPM-SDE augmented-history transition.

9 Conclusion

The small RL checkpoint should be viewed as a cheap instrument that discovers a policy-improvement direction, not as an absolute teacher for the larger model. The general transferable signal is the conditional kernel ratio $\mathrm{d}K_{S_R}/\mathrm{d}K_{S_0}$. Its geometric tilt of the large reference kernel is an exact KL-regularized objective without any Gaussian assumption. Shared Gaussian SDE and DPM-SDE kernels make the target computationally simple: the donor transition-mean shift is added to the large reference mean, and the resulting training loss is a covariance-weighted residual MSE. Continuous SDEs give the same local drift target through Girsanov, while the exact global path tilt additionally contains a continuation-value correction.

For ODE flow models, transition ratios are not defined. Nevertheless, the same velocity arithmetic is a principled rescaled small-noise control limit and a valid deterministic field construction. Flow-map CFM and endpoint density-ratio methods provide alternative deterministic and distributional targets when the Eulerian donor shift is unreliable. The practical research question is therefore not whether a 9B model is a good teacher for a 32B model, but whether the *change discovered by 9B RL* is stable and useful on the 32B model's own states.

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